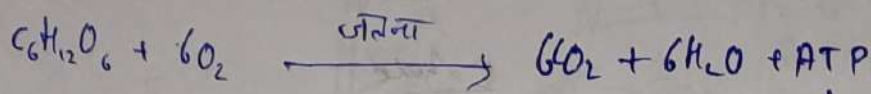


Respiration System

Respiration $\Rightarrow$ Bio-chemical



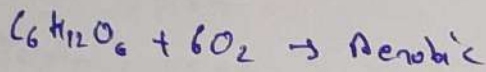
cell $\rightarrow$ Mitochondria
 $\hookrightarrow$ cellular respiration

Energy currency of cell

Respiration

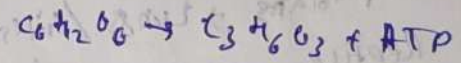
Aerobic

$\hookrightarrow O_2$



Anaerobic

$\hookrightarrow O_2 \times$



Ethanol

cellular respiration / Aerobic respiration

$\hookrightarrow$ site

$\hookrightarrow$ mitochondria

$\hookrightarrow$ cytoplasm

$\hookrightarrow$ 38 molecule (ATP)

Ex Human, Bird, Animals.

Anaerobic respiration:

$\rightarrow$ site

$\hookrightarrow$ cytoplasm

$\hookrightarrow$ muscle cell, seed.

Breathing

↳ Air → Inhale

Air → Exhale

Inhale

$N_2 \rightarrow 78\%$

$O_2 \rightarrow 21\%$

$CO_2 \rightarrow 0.04\%$

Other $\rightarrow 2\%$

Exhale

$N_2 \rightarrow 78\%$

$O_2 \rightarrow 16\%$

$CO_2 \rightarrow 4-5\%$

Other $\rightarrow 2\%$

Respiratory Organs:

Fish $\rightarrow$ gills

Bird $\rightarrow$ lungs

Amphibians $\rightarrow$ Lungs / gills / skin

Reptiles $\rightarrow$ lungs (hibernation)

mammals $\rightarrow$ Lungs

Respiratory System

Nasal cavity $\rightarrow$ air enters

Pharynx $\rightarrow$ air enters the food pipe with it

Larynx $\rightarrow$ voice box

Trachea $\rightarrow$ wind pipe

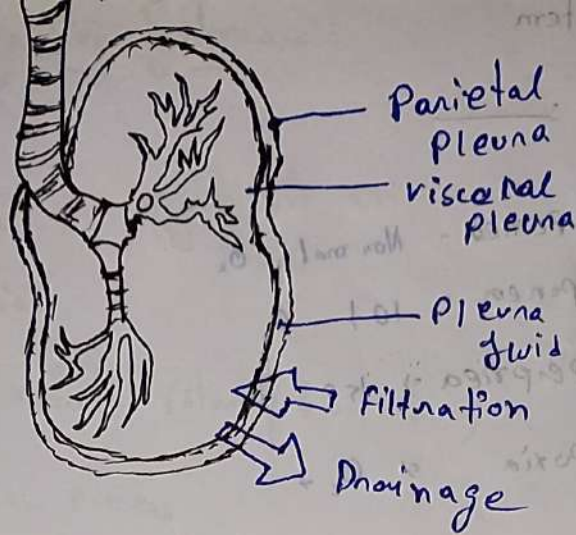
1st division of trachea $\rightarrow$ Bronchi

Bronchioles $\rightarrow$ Tree like structure

Alveoli $\rightarrow$ grapes like structure

unit of respiratory system

Alveoli.



action _____

- Blood Purify

Lungs

Colour $\rightarrow$ pink
 Weight $\rightarrow$ 1.5 kg
 Smelling $\rightarrow$ sweet

Larynx :-

Boys $\rightarrow$ 20-25 mm

Girls $\rightarrow$ 15-20 mm

Adam's Apple $\rightarrow$ Boys

Respiratory Rate :

Children $\rightarrow$ 20-30/m.

Adult $\rightarrow$ 12-18/m.

instrument $\rightarrow$ to measure Spirometer

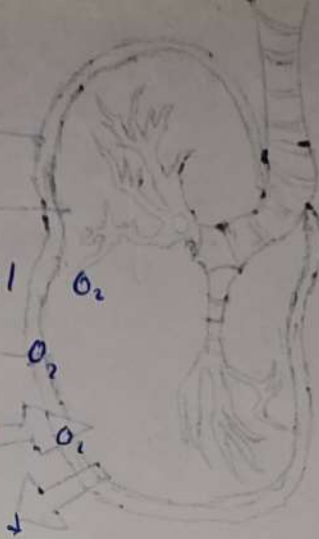
Brain $\rightarrow$ Pons Vannioli $\rightarrow$ Respiration

Respiratory system $\rightarrow$ medulla oblongata

Respiratory System

- Disease

Eupnea - Normal O_2
 Hypnea $\rightarrow 10 \downarrow O_2$
 Hyperpnea $\rightarrow 15 \uparrow O_2$
 Hypoxia $\rightarrow O_2 \downarrow$



Anoxia $\rightarrow O_2 (x)$

Emphysema $\rightarrow$ Smoke $\rightarrow$ Alveolies $\rightarrow$ Pores

pneumonia $\rightarrow O_2 \downarrow$

Asthma $\rightarrow$ fungi $\rightarrow$ R.S.

Nasal cavity

$\rightarrow$ Pollen grains

$\rightarrow$ जलवायु, धूलिका

T.B $\rightarrow$ Tuberculosis

$\rightarrow$ Bacteria

(MTB)

Bch, Vaccine

White Lungs $\rightarrow$ cotton

Black Lungs $\rightarrow$ cement/coal